H-3 P2400108 LeX

(1) $u(t)-u(t-3)=\begin{cases}1 & t\in[0,10] \\ 0 & \text{other}\end{cases}$

$E=\int_{-\infty}^{+\infty}x^2(t)dt=\int_0^{10}A^2dt=10A^2$

(b) $x(t)=\text{rect}(t)\cos(2\pi t)$

$E=\int_{-\infty}^{+\infty}[\text{rect}(t)\cos(2\pi t)]^2dt=\int_{-0.5}^{0.5}\cos^2(2\pi t)dt$

$\cos^2(2\pi t)=\frac{1+\cos(4\pi t)}{2}$

$E=\int_{-0.5}^{0.5}\frac{1+\cos(4\pi t)}{2}dt=\frac{1}{2}\int_{-0.5}^{0.5}1\,dt+\frac{1}{2}\int_{-0.5}^{0.5}\cos(4\pi t)dt$

$E=\frac{1}{2}$

2、$P=\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}|x(t)|^2dt$

(a) $x(t)=u(t)$

$P=\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}u^2(t)dt=\lim_{T\to\infty}\frac{1}{2T}\int_0^T1\,dt=\frac{1}{2}$

(b) $x(t)=A\cos(2\pi f_0t)$

$P=\frac{1}{T_0}\int_{T_0}x^2(t)dt$

$\frac{1}{T_0}\int_{T_0}^{T_0}[A\cos(2\pi f_0t)]^2dt=\frac{A^2}{T_0}\int_{T_0}^{T_0}\cos^2(2\pi f_0t)dt$

$P=\frac{A^2}{T_0}\left(\frac{1}{2}\int_0^{T_0}1\,dt+\frac{1}{2}\int_0^{T_0}\cos(4\pi f_0t)dt\right)$

$P=\frac{A^2}{2}$

3、linear: $y(aX_1(t)+bX_2(t))=ay(X_1(t))+by(X_2(t))$

$y(t)=[aX_1(t)+bX_2(t)]_{t\to\frac{t}{2}}=aX_1(\frac{t}{2})+bX_2(\frac{t}{2})$

$aX_1(t)+bY_2(t)=aX_1(\frac{t}{2})+bX_2(\frac{t}{2})$

$y'(t)=aX_1(t)+bY_2(t)$

time variant: $y(t-\tau)=X(\frac{t-\tau}{2})\neq$

$y(t)=X(\frac{t}{2}-\tau)$

non-causal: $y(t)=X(\frac{t}{2})$

when $t<0$, $\frac{t}{2}>t$

non-causal

4. $y(t)=\cos[x(t)]$

nonlinear: $x_1(t)+x_2(t) \Rightarrow y''(t)=\cos[\pi]=-1$

$y_1(t)+y_2(t)=-1-1=0 \neq -1$.

time invariant: $y_\tau(t)=\cos[x(t-\tau)]$ ‖

$y(t-\tau)=\cos[x(t-\tau)]$ ✓

BIBO: $\cos[\,] \leq 1$ $|\cos[x(t)]| \leq 1$ stable.

memoryless: $y(t)=\cos[x(t)]$ $t \to t$ only depend

Casual ⇓ causal.

5. $y(t)=t^2 x(t-1)$

time variant: $y_\tau(t)=t^2 x(t-1-\tau)$

$y(t-\tau)=(t-\tau)^2 x((t-\tau)-1)=(t-\tau)^2 x(t-1-\tau)$

$t^2 \neq (t-\tau)^2$ $(\tau \neq 0)$

BIBO: $|x(t)|=1 \leq M=1$ $y(t)=t^2$.

$t \to \infty$ $t^2 \to \infty$ not stable.

casual: $y(t_0)=t_0^2 x(t_0-1)$

memory $t_0 \to t_0-1$.

non invertible: $x_1(t) \neq x_2(t)$, $y_1(t) \neq y_2(t)$

$x_1(t)=1$

$x_2(t)=1$

6. $y(t)=t\,x(t)$

(a) $y(ax_1+bx_2)=t(ax_1+bx_2)=atx_1+btx_2$

$=ay_1+by_2$.

linear.

time-invariance

$x(t-\tau) \to tx(t-\tau) \neq (t-\tau)x(t-\tau)$

BIBO $|x(t)| \leq 1$, $y(t)=t$

$t \to \infty$. $y(t) \to \infty$

not stake

Causality: $y(t)=t\,x(t)$

no

memory no

Invertibility $x(t)=\frac{y(t)}{t}$.

✓ yes

(b) $y(t)=\int_{t_0}^{t} x(\tau)\,d\tau$

linear: $\int_{t_0}^{t}[ax_1(\tau)+bx_2(\tau)]\,d\tau=ay_1(t)+by_2(t)$

time-invariance: $x(t-\tau)$

$\int_{t_0}^{t} x(\tau'-\tau)\,d\tau'=\int_{t_0-\tau}^{t-\tau} x(\sigma)\,d\sigma$.

Yes.

BIBO:

$|x(t)| \leq 1$, $y(t)=t-t_0$ $t \to \infty$ $y(t) \to \infty$

no stable.

Casual: $t \to t-t_0$ Yes

invertibility: $\frac{dy(t)}{dt}=x(t)$

Date / /

No.

7. $x(t)=e^{at}u(t)$

(1) $a<0$, $x(t)=e^{-bt}u(t)=\begin{cases}e^{-bt}, & t\geq 0\\ 0, & t<0\end{cases}$

$E=\int_{-\infty}^{+\infty}x^2(t)dt=\int_0^{+\infty}(e^{-bt})^2dt=\int_0^{+\infty}e^{-2bt}dt$

$\int_0^{+\infty}e^{-2bt}dt=\frac{e^{-2bt}}{-2b}\Big|_0^{\infty}=\frac{1}{2b}=\frac{1}{-2a}$

$a<0$, $E<+\infty$.

$P=\lim_{t\to\infty}\frac{1}{2T}\int_{-T}^{T}x^2(t)dt=\lim_{T\to\infty}\frac{1}{2T}\frac{1}{2b}=0$.

$a<0$, energy signal.

(2) $a>0$, $x(t)=e^{at}u(t)\begin{cases}e^{at} & t\geq 0\\ 0 & t<0\end{cases}$

$E=\int_{-\infty}^{\infty}x^2(t)dt=\int_0^{+\infty}e^{2at}dt$

$E=\infty$

$P=\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}x^2(t)dt=\lim_{T\to\infty}\frac{1}{2T}\frac{e^{2aT}-1}{2a}$

$a>0$, $e^{2aT}\to\infty$

$P=\infty$

not energy signal not power signal.